

Research Stay Report: Big or Small: The Effect of Size on Group Dynamics in a Crowded Environment

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Abstract

Humans, ants, and active particles all form pathways in their environment to enable more efficient navigation. Active particles have been shown to create and reuse tunnels in a sea of passive obstacles, leading to the formation of large, organized structures (clusters). In this project, we aim to elucidate how the size and dynamics of these clusters depend on the active particle and obstacle size using numerical simulations. We map the phase space by systematically varying the particle radius and obstacle density, revealing a strong dependence of maximum cluster size and rotational stability on these parameters.

1 Theoretical Background

1.1 Brownian Motion

At the microscopic scale, particles suspended in a fluid undergo continuous, random motion due to thermal agitation and collisions with the surrounding fluid molecules. This phenomenon is known as Brownian motion [1]. Initially, at very short time scales, the motion of a particle is ballistic, governed by its inertia, where its mean square displacement (MSD) grows quadratically with time ($MSD(\Delta t) \propto \Delta t^2$) [1]. However, as the particle experiences numerous collisions, its velocity becomes randomized. Beyond a characteristic momentum relaxation time, the motion transitions into the diffusive regime, where the MSD grows linearly with time ($MSD(\Delta t) \propto \Delta t$) [1]. This diffusive behavior is a hallmark of passive colloidal particles in thermal equilibrium.

1.2 Active Particles and Active Brownian Motion

In contrast to passive particles, active particles are capable of continuously converting stored or ambient energy into directed mechanical motion [2, 3]. When this self-propulsion is combined with rotational diffusion (a process that even passive particles undergo, although it is inconsequential since they do not have a self-propulsion speed), the resulting dynamics are termed active Brownian motion. Active particles, such as Janus colloids, can be propelled by various mechanisms; for instance, light absorption on a coated hemisphere can induce local heating and asymmetric fluid demixing, generating a propelling force [2].

The trajectories of active particles are characterized by persistence over short time scales. We must distinguish between the scales of motion: at a scale shorter than the momentum relaxation time, all particles behave ballistically. Beyond this scale, passive particles are always diffusive. For active Brownian particles, the behavior is diffusive at short scales (larger than the momentum relaxation time), ballistic at intermediate scales, and eventually diffusive again at long scales with enhanced diffusion. The dynamics of a self-propelled active particle i with constant speed v and orientation angle ϕ_i can be described by the overdamped Langevin equations:

$$\dot{\mathbf{x}}_i = -\frac{1}{\gamma_t} \nabla_{\mathbf{x}} V_i + v \hat{\mathbf{n}}_i + \sqrt{2D_t} \boldsymbol{\xi}_i, \quad (1)$$

$$\dot{\phi}_i = \frac{1}{\gamma_r} \Omega_i + \sqrt{2D_r} \xi_{r,i}, \quad (2)$$

where $\mathbf{x}_i$ is the particle position, $\hat{\mathbf{n}}_i = [\cos(\phi_i), \sin(\phi_i)]$ is the orientation vector, γ_t and γ_r are the translational and rotational friction coefficients, D_t and D_r are the translational and rotational diffusion coefficients, and $\boldsymbol{\xi}_i$ and $\xi_{r,i}$ represent translational and rotational white noise, respectively. The terms V_i and Ω_i account for the interaction potential and environmental torque, respectively [3].

1.3 Interactions and Potentials in Active Systems

The emergent collective behaviors of active particles, such as the formation of dynamic clusters, depend fundamentally on the forces and potentials they experience.

- **Particle-Particle Interactions:** Active particles interact with one another, and these interactions can be mathematically modeled to capture their behavior using a Lennard-Jones potential V_{ij} between parti-

cles i and j separated by a distance r_{ij} :

$$V_{ij}(r_{ij}) = 4\epsilon_{LJ} \left[\left(\frac{\sigma_{LJ}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{LJ}}{r_{ij}} \right)^6 \right], \quad (3)$$

where ϵ_{LJ} is the depth of the potential well and σ_{LJ} is the potential width [2, 3].

- **Environmental Feedback and Aligning Torques:** When active particles navigate crowded environments containing passive obstacles, they experience asymmetric interactions. For example, the asymmetric demixing of a fluid around a Janus particle near an obstacle can induce a deterministic aligning torque [2, 3]. This effective torque Ω_i acting on an active particle i due to N_p surrounding passive particles can be expressed as:

$$\Omega_i = -4\Omega_0 R^2 [\hat{\mathbf{n}}_i, 0] \times \sum_{j=1}^{N_p} \nabla_{[\mathbf{x}_i, 0]} \frac{e^{-\kappa r_{ij}}}{r_{ij}}, \quad (4)$$

where Ω_0 is the torque strength, R is the particle radius, and κ is the screening length [3]. This torque effectively steers the active particle away from regions of high obstacle density and towards voids or pre-existing paths. This environmental feedback is crucial for phenomena such as stigmergy (indirect coordination via shared environmental memory) [2] and the stabilization of collective rotational order (spinning) in living crystals [3].

1.4 Energy Scaling and Phenomenological Collapse

A critical aspect of modeling active particles of varying sizes is understanding that we are primarily interested in the clustering behavior in the absence of passive particles. Therefore, we change the potential well depth to remove this effect [3]. To systematically compare the clustering dynamics of particles with different radii (R), we apply a scaling law to the Lennard-Jones energy depth (ϵ_{LJ}):

$$\epsilon(R) = \epsilon_{LJ} \left(\frac{R}{R_0} \right)^\alpha, \quad (5)$$

where $R_0 = 2.385\mu\text{m}$ is a reference radius from the experimental system and α is a scaling exponent. We cannot definitively state how the interaction energy scales with particle size in reality. However, our numerical analysis reveals that applying an exponent of $\alpha = 1.8$ provides the optimal phenomenological collapse of the cluster size (C_{max}) curves to reproduce structurally equivalent clustering behaviors.

1.5 Translational and Rotational Dynamics of Living Crystals

To properly characterize the collective behaviors of the living crystals, specifically their translational and rotational dynamics, we rely on a set of precise observables [3]. The rotational dynamics are fundamentally driven by the torque exerted on the crystal's center of mass (COM) by the self-propulsion of its constituent particles, defined as:

$$M = \gamma_t v \sum_{i=1}^N \mathbf{r}_i \times \hat{\mathbf{n}}_i, \quad (6)$$

where $\mathbf{r}_i$ is the vector connecting the COM to particle i . The total torque M can be further divided into contributions M_P and M_C from perimeter and core particles, respectively.

To evaluate the rotational order and the alignment of the particles relative to the COM (as analyzed in the probability distribution $P(\Phi)$), we define the rotational order parameter, Φ :

$$\Phi = \frac{1}{N} \sum_{i=1}^N (\hat{\mathbf{r}}_i \times \hat{\mathbf{n}}_i) \cdot \hat{\mathbf{u}}_z, \quad (7)$$

where $\hat{\mathbf{r}}_i$ is the unit vector of $\mathbf{r}_i$, and $\hat{\mathbf{u}}_z$ is the unit vector orthogonal to the plane of the crystal [3]. This parameter provides a crystal-size-independent measure where $\Phi \in [-1, 1]$, with -1 and 1 indicating alignment compatible with clockwise and counterclockwise rotations, respectively, and 0 representing parallel or anti-parallel alignment to the radial direction.

Additionally, to quantify the overall tendency of a living crystal to expand or contract, we introduce the radial moment G :

$$G = \sum_{i=1}^N G_i = \gamma_t v \sum_{i=1}^N \mathbf{r}_i \cdot \hat{\mathbf{n}}_i. \quad (8)$$

A positive G indicates that particles tend to self-propel away from the COM (expansion), while a negative G points towards contraction [3].

2 Methodology

2.1 Numerical Simulation Framework

To simulate the behavior of active particles in a crowded environment of passive obstacles, is used a C++ script that implements a two-dimensional,

overdamped Langevin dynamics simulation. The simulation operates within a square domain with periodic boundary conditions, utilizing a discrete time step Δt and an Euler integration scheme to iteratively update particle positions and orientations.

The script models N_a active particles and N_p passive particles. The core of the numerical engine relies on resolving two main types of interactions using a highly optimized Verlet neighbor list algorithm:

1. **Translational Forces (Lennard-Jones):** All particles interact via a pairwise Lennard-Jones potential. For active-active interactions, both the repulsive and attractive components of the potential are calculated up to a specified cutoff distance of 5 particle diameters. For interactions involving passive particles, only the repulsive core is considered to simulate steric hindrance, utilizing a cutoff of 1 particle diameter. An iterative position relaxation algorithm is applied during initialization to resolve unphysical overlaps by gradually increasing the radius of the passive particles, which consequently affects their density.
2. **Rotational Torques (Environmental Feedback):** The script computes an effective torque Ω_i for each active particle based on its proximity to surrounding passive obstacles up to a distance of 4 particle diameters. This torque alters the orientation angle ϕ_i of the active particle, explicitly steering it away from regions of high passive particle density.

2.2 Data Analysis and Cluster Identification

To automate the massive simulation campaign, a Python-based parallel execution script was utilized, running 15 independent replicas per condition for a total of 3000 seconds to ensure robust statistical significance. Data extraction relied on spatial analysis using k-dimensional trees (`ckDTree`) with a cutoff distance of $2.2R$ and graph-based connected components to identify interacting networks of particles. A valid group is defined as a connected component comprising at least 3 active particles.

To guarantee that the system had sufficient time to move away from the random initialization before extracting structural metrics, the first 2000 seconds of each simulation replica were discarded. For the rotational order analysis, the algorithm specifically isolates the largest cluster (requiring a minimum of 7 particles), carefully unwraps the periodic boundary conditions to accurately calculate the center of mass, and computes the probability distributions of the rotational order parameter, $P(\Phi)$. Furthermore, to visually

validate the physical mechanisms and clustering dynamics, representative replicas—those whose behavior was closest to the global statistical mean—were automatically identified and rendered into MP4/GIF animation videos.

3 Results and Discussion

To analyze the system’s behavior, we explored three different active particle radii ($R = 0.4R_0$, $R = 1.0R_0$, and $R = 1.6R_0$, where $R_0 = 2.385\mu\text{m}$ corresponds to the reference particle radius in the experimental setup) against a range of passive particle (obstacle) densities, denoted as ρ_p .

3.1 Temporal Evolution and Initialization

Before analyzing the maximum cluster sizes, we monitored the temporal evolution of the cluster size (C_{max} vs t) from $t = 0$ to $t = 3000$ seconds for various densities (Figure 1).

- For small ($R = 0.4R_0$) and medium ($R = 1.0R_0$) radii, the cluster size undergoes rapid initial growth followed by stabilization (a plateau) for some densities.
- For the largest radius ($R = 1.6R_0$), smaller particles actually experience larger fluctuations in terms of translational and rotational diffusion, contrary to some assumptions.

Importantly, across all three radii, there is at least one density curve where the cluster size is continuously increasing, even beyond 3000s. Therefore, the system is not entirely in a steady state, but the chosen timeframe allows the system to evolve sufficiently away from the random initialization state for our analysis.

Temporal evolution of cluster size for steady-state identification

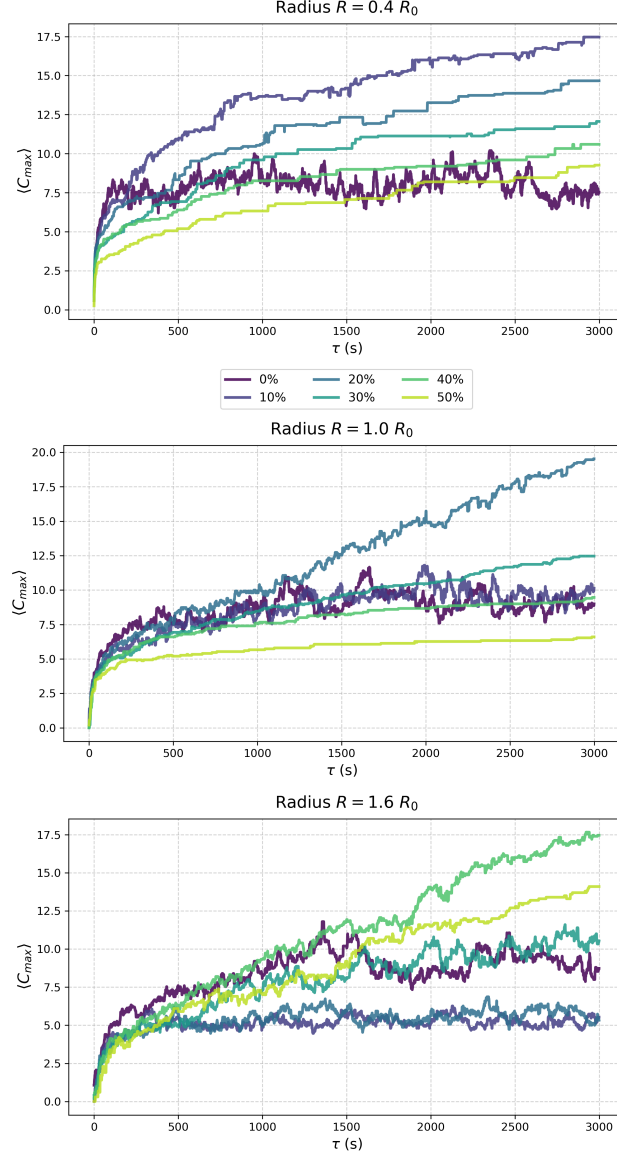


Figure 1: Temporal evolution of the maximum cluster size (C_{max}) to verify the stabilization away from initialization across different particle radii and obstacle densities.

3.2 Effect of Obstacle Density on Cluster Size

Figure 2 reveals a strong non-linear dependence between the maximum cluster size (C_{max}) and the passive particle density (ρ_p), as well as the active

particle radius. The structural data extracted from the post-initialization regime demonstrates distinctly different peaks for varying radii:

- **Small particles ($R = 0.4R_0$):** Exhibit a very pronounced aggregation peak at low densities ($\rho_p \approx 5\% - 10\%$). At higher densities, the trend breaks down, and the cluster size drops drastically. This occurs because the small particles become physically trapped or caged within the dense passive matrix, restricting their mobility and preventing them from navigating the medium to aggregate with other active particles.
- **Medium particles ($R = 1.0R_0$):** The peak maximum size shifts towards intermediate densities ($\rho_p \approx 15\% - 20\%$) and its maximum magnitude is lower ($C_{max} \approx 17$).
- **Large particles ($R = 1.6R_0$):** Require a much denser medium to maximize their cluster size, showing C_{max} peaks in regions of high obstacle density ($\rho_p > 35\%$).

This phenomenological observation points to a complex dynamic interplay where many effects take place in reality, including the balance of noise and self-propulsion, the closing of pathways, the stability provided by the torque, and the feedback provided by the environment. The channels (or tunnels) that open up within the passive matrix actively assist larger particles in navigating and aggregating into stable clusters. Conversely, these same dense environments trap smaller particles. The stability of the pathways depends strongly on the size of the particles; smaller particles have a larger translational diffusion (D_t), which allows the passive particles to quickly fill and close the pathways behind them.

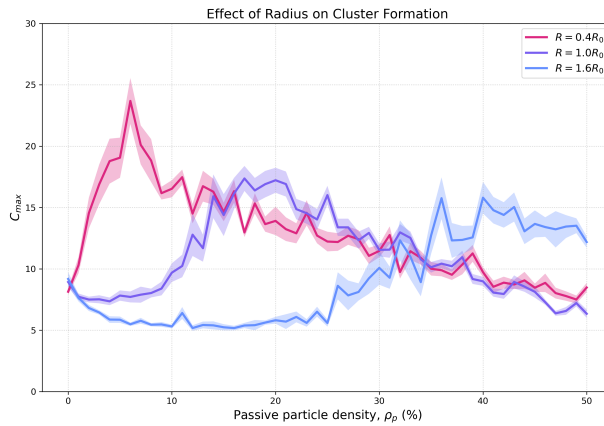


Figure 2: Effect of passive particle density (ρ_p) on the maximum cluster size (C_{max}) for different active particle radii.

3.3 Onset of Spontaneous Rotation and Dynamics

To test the hypothesis of rotational stability and understand the dynamic mechanisms at play, we analyzed the probability distribution of the rotational order parameter, $P(\Phi)$, as the density varied (Figure 3). These results, combined with the visual confirmation from the rendered representative animations, provide a comprehensive view of the system’s dynamics:

- **Low-density regime ($\rho_p = 0\%$ to 10%):** Regardless of particle size, the distributions are unimodal and centered at $\Phi = 0$. In this regime, cluster formation and rotation act via different mechanisms. For small particles ($R = 0.4R_0$), large clusters form, but the environment does not provide enough feedback to overcome their incredibly high noise to induce rotation. For larger radii, there is not constant enough feedback from the environment, so alignment within the clusters never builds up.
- **Transition at high density ($\rho_p \geq 20\%$):** As the number of obstacles increases, a clear size-dependent symmetry breaking emerges:
 - Small particles ($R = 0.4R_0$) maintain a sharply unimodal and narrow distribution, retaining their rotational stability in the sense that they are rotating very quickly—more so than the environment can order them. However, as noted previously, this is largely due to their localized trapping within the dense matrix.
 - Conversely, medium and large particles ($R = 1.0R_0$ and $R = 1.6R_0$) undergo a transition towards a much broader, bimodal distribution (peaks developing towards ± 0.5). This bimodality is clear evidence that larger clusters experience the onset of spontaneous rotation. The dynamic interaction with the dense passive environment induces collective structural rotations (spinning) within the formed pathways, distinguishing the mechanism of channel creation from the mechanism of rotation itself.

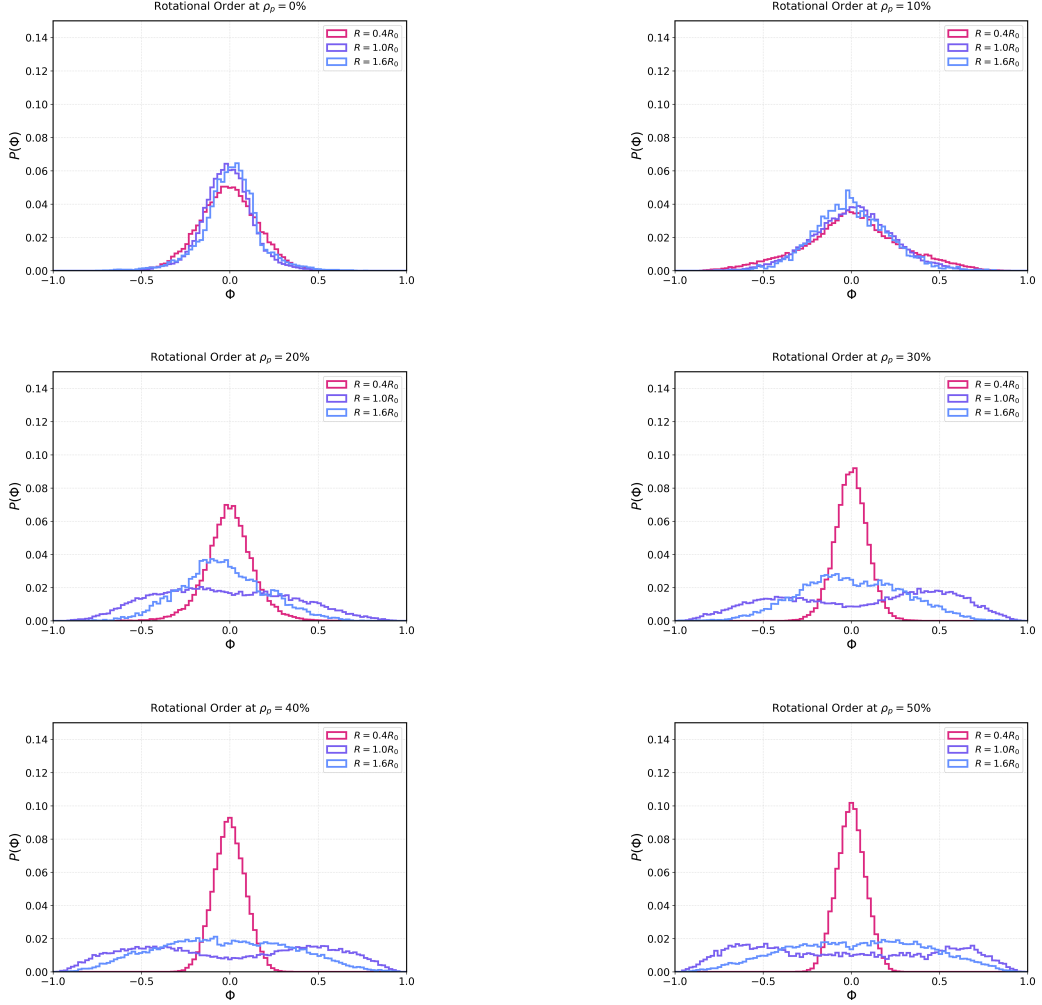


Figure 3: Probability distributions of the rotational order parameter $P(\Phi)$ for densities $\rho_p \in \{0\%, 10\%, 20\%, 30\%, 40\%, 50\%\}$. Notice the transition from unimodal to bimodal distributions for larger radii at high densities.

To further quantify this symmetry breaking across the entire spectrum, we mapped the peak values of the rotational order parameter (Φ_{peak}) against the passive particle density, yielding the bifurcation diagram shown in Figure 4. This representation clearly delineates the phase space of the system's rotational dynamics:

- **Small particles ($R = 0.4R_0$):** The peak remains firmly locked at $\Phi_{peak} = 0$ across all tested densities, as they do not spontaneously transition into a spinning state.
- **Medium particles ($R = 1.0R_0$):** Exhibit a distinct, classic pitchfork-

like bifurcation at a critical density threshold of $\rho_p \approx 20\%$. Beyond this point, the system spontaneously breaks symmetry, adopting a tendency to spin where the peaks stabilize at approximately ± 0.5 . This is not a fixed spinning state, since the cluster has no preference for which way it spins and in fact changes rotation periodically.

- **Large particles ($R = 1.6R_0$):** Also undergo a symmetry breaking (loss of rotational stability), but the bifurcation is significantly noisier. We report this phenomenon, though determining the exact causes of this noise requires further analysis beyond the scope of this phenomenological observation.

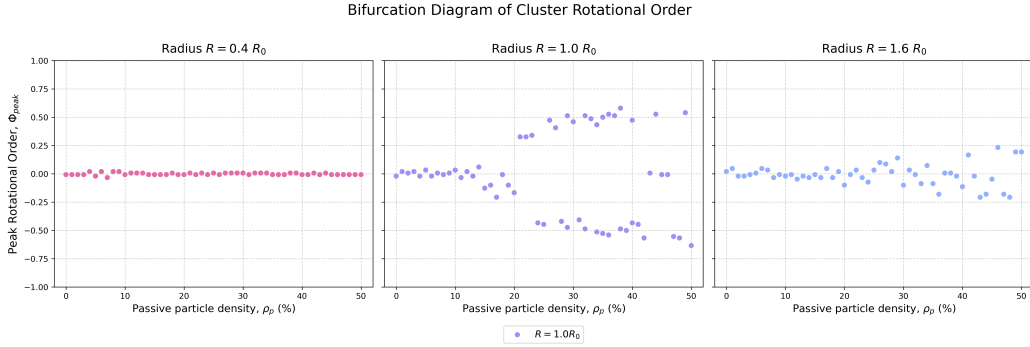


Figure 4: Bifurcation diagram tracking the peak of the rotational order parameter (Φ_{peak}) as a function of passive density (ρ_p) for three particle radii. Notice the clear pitchfork bifurcation for $R = 1.0R_0$ starting at $\rho_p \approx 20\%$.

4 Conclusions

- Our simulations confirm that the active particle size dictates the optimal obstacle density required for pathway formation and large cluster assembly. The channels formed in the passive matrix assist large particles in aggregating in highly populated environments, while small particles become trapped as density increases.
- The hypothesis regarding rotational dynamics was validated: small particles form clusters that maintain unimodal rotational distributions even in dense media. In contrast, an increase in particle radius induces a transition towards the onset of spontaneous rotation. The bifurcation analysis pinpoints a critical density threshold ($\rho_p \approx 20\%$ for medium particles) where spontaneous symmetry breaking occurs, leading to bimodal rotational tendencies.

- The phenomenological analysis demonstrated that applying an exponent of $\alpha = 1.8$ acts as a numerical tool to collapse the cluster size distributions across varying particle radii.

References

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